MILITARY SPECIFICATION

GAGE, FUEL QUANTITY, CAPACITOR TYPE, COMPENSATED,
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-G-7818, dated 26 November 1951, and was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

Paragraph 3.2.2 Delete the last sentence and substitute the following:

"Insofar as practicable, pivots, bearings, and gears shall neither bind nor shake and shall be frictionless."

Paragraph 3.2.3 Fire Hazard: Add the following subparagraph:

"3.2.3.1 The design of the circuit shall be such that the current in the leads running to the sensing electrodes of tank units and compensator sensing units shall in no case exceed 0.2 ampere. This requirement shall apply both to normal operating conditions and to any combination of the following abnormal conditions:

(a) Short circuiting of any combination of electrodes in the tank unit or compensator sensing unit.

(b) Short circuiting of any capacitor or combination of capacitors elsewhere in the bridge circuit.

(c) All switches and other adjustments in the bridge circuit set to any desired position."

Paragraph 3.2.4 General: At the end of the first sentence add the following phrase:

"including turbo-prop applications."

Paragraph 3.5.1 Totalizer: Delete the last sentence.

Paragraph 3.5.2: At the end of the first sentence add the following phrase:

"unless approved by the procuring agency."

Paragraph 3.5.3 Tank Units: Delete in its entirety and substitute the following paragraph:

"3.5.3 Tank Units.- The basic tank unit shall consist of a suitable capacitative arrangement of metallic conductors such as a pair of concentric cylinders electrically insulated from each other and forming a three-wire capacitor. The design shall be such that the tank unit can be disassembled and reassembled without the use of special tools, except as approved by the procuring agency. It shall be so constructed as to permit fuel into which the unit has been immersed to rise freely to its natural level between the conductors, so that up to this level, the fuel will serve as the dielectric substance.

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Above the level of the fuel, a mixture of air and vapor will be the dielectric substance. Unless otherwise approved, non-sensing portions of the sensing element shall not exceed the following dimensions: 0.600 inch at the flange end and 0.400 inch at the opposite end for externally mounted tank units, except that 0.850 inch shall be allowed for the flanged end only for tank units including the compensator capacitors, and 0.400 inch at both ends for internally mounted tank units. It shall be so constructed as to preclude the formation of deposits and sediments between the conductors. The minimum allowable distance between the conductors shall be 0.125 inch. The tank unit shall be designed to make electrical connections readily accessible for testing the tank unit and making calibration adjustments using a substitute precision capacitance. The materials used in the construction of the tank units shall be impervious to aircraft fuels."

Paragraph 3.5.4 Intermediate Device: Add the following sentence between the first and second sentences:

"The intermediate device shall be designed to provide readily accessible electrical connections for testing the gage system and making calibration adjustments using substitute precision capacitors."

Paragraph 3.5.5.1: Delete the third sentence.

FIGURE 1: Delete "LENGTH OF CASE SHALL NOT EXCEED 4.25 INCHES."

FIGURE 2: Make the following changes:

(a) Opposite "TOLERANCES" delete"ANGLES ±0°15'"

(b) Delete the following note: "NOTE: LENGTH OF CASE SHALL NOT EXCEED 5 INCHES"

(c) In the marking data for the "MAIN DIAL," under the second column, change as follows: (1) Opposite "FUEL QUANTITY," delete ".140" and substitute ".125"; (2) Opposite "LBS X," delete ".100" and substitute ".062"

(d) In the marking data for the "SUB-DIAL," in the first column, delete "X100" and substitute "X-100, OR X-1000 AS APPLICABLE"

(e) Change sub-dial circle diameter from "0.718 ±.010" to "0.583 ±.010."

FIGURE 3: Make the following changes:

(a) Delete the following note: "NOTE: LENGTH OF CASE SHALL NOT EXCEED 4.25 INCHES"

(b) Opposite "TOLERANCES" delete "ANGLES ±0°-15'"

Paragraph 3.5.5.2 Indicator Case: Add the following sentences at the end of the paragraph:

"The case length shall not exceed 4.25 inches for nonsensitive indicators and 5.00 inches for sensitive indicators. The length limitations may be modified only with the specific approval of the procuring agency."

Paragraph 3.5.5.5.2 Dial Scale: In the fifth and seventh lines, delete "radius" and substitute "angle."

Paragraph 3.5.5.5.3: In item (c) delete "usable" and add the following at the end of the sentence: "By tank volume is meant that part of the total volume which is measured by the gage."

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Paragraph 3.5.5.5.4 Dial Markings: Add the following sentence between the third and fourth sentences: "The graduation markings for the sub-dials on figures 2 and 3 shall be linear within ±1°."

Paragraph 3.9 Damping: Delete in its entirety and substitute the following paragraph:

"3.9 Pointer Overshoot.- The gage shall be designed to provide steady indication of the pointer. In approaching null the pointer shall not overshoot the final position more than one time and shall return to the final position within 2 seconds."

Paragraph 3.11 Gage Failure Indication: Delete in its entirety and substitute the following paragraph:

"3.11 Gage Failure Indication.- The gage shall be provided with a readily accessible terminal connection for use with a remote test switch to indicate failure of gage operation. Operation of the external test switch shall cause the indicator pointer to rotate counterclockwise from a balance point on any portion of the dial."

Paragraph 3.16.4 Wiring Diagram: Delete the entire paragraph.

Paragraph 4.2.4.1 Test Instrument: Delete the entire paragraph and substitute the following:

"4.2.4.1 Test Instrument.- Whenever a master test instrument is specified for tests of gages, it shall be in the form of a precision variable capacitor or precision fixed capacitors. The accuracy of the precision capacitors, fixed or variable, at 25°C, shall be 0.2 percent for capacitances in excess of 50 uuf and 0.1 uuf for capacitances less than 50 uuf. For tests of totalizers the master test instrument shall consist of a suitable number of gages, or equivalent input devices, which have been calibrated to an accuracy of 0.2 percent."

Paragraph 4.2.4.2 Capacitance Bridge: Delete the second sentence and substitute the following: "The accuracy of the bridge at an ambient temperature of 25°C shall be 0.2 percent for capacitances in excess of 50 uuf and 0.1 uuf for capacitances less than 50 uuf.

Paragraph 4.2.6: Delete the entire paragraph and substitute the following:

"4.2.6 In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement."

Paragraph 4.3.1 Sampling Instructions: In the first sentence add "two" immediately preceding "totalizers."

Paragraph 4.3.1.2 Totalizer: Delete the paragraph in its entirety and substitute the following:

"4.3.1.2 Totalizer.- Totalizers submitted for qualification approval shall be designed to combine the indications of three gages corresponding to relative tank volumes of 6:3:1. The requirements for the totalizer indicator shall be the same as those for the gage indicator. Totalizers shall not be submitted for qualification unless approval has been previously authorized for individual gages associated with the totalizer, or the individual gages are submitted jointly with the totalizers for qualification."

Paragraph 4.3.1.2.1 Gage Outputs for Totalizer: Delete the second and third sentences.

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Paragraph 4.3.1.2.2 Totalizer Inputs: Delete the paragraph in its entirety and substitute the following:

"4.3.1.2.2 Totalizer Inputs.- The manufacturer shall furnish with the totalizers a set of three gages, provided with suitable calibration charts to serve as inputs for testing the totalizers. In lieu of the three gages, the manufacturer may submit artificial devices simulating the gages. If artificial devices are submitted, the manufacturer shall provide sufficient data to verify that such devices are equivalent to the respective individual gages."

TABLE I Characterization of Tank Units for Qualification Tests:. Under the fourth column in the sixteenth line change "22.24" to read "25.24." Under the sixth column on the last line, change "0" to read "2.4."

Paragraph 4.3.1.4 Calibration Data for Qualification Tests: Delete subparagraph "(c)" in its entirety.

Paragraph 4.4.1 Calibration Data for Inspection Tests: Delete item (h) and substitute the following:

"(h) Relative positions of tank units corresponding to locations in the airplane."

Paragraph 4.4.2 Individual Tests: Add the following at end of paragraph: "unless otherwise specified."

Paragraph 4.4.2.2 Indicator Operation at Room Temperature: Delete item (c) in its entirety and substitute:

"(c) That, where applicable, the output signal at each major graduation is within ±1/2 percent of the fixed design value at full scale. This test is required only for those indicators providing signals for totalizers."

In item (e) delete the clause following "the indicator."

TABLE III Gage System Calibration Data: Change the heading of the fifth column to read "Output Signal Where Applicable."

Revise note to read as follows:

"NOTE: The last four columns of this table shall be extended to include capacitance values and totalizer output signals up to and including the highest major dial graduation. The capacitance values listed in the extended portion of the second column are to be interpreted as applicable to those tests on which master test instruments are connected in lieu of the tank units."

TABLE IV Totalizing Gage Calibration Data (Intermediate Device and Indicator): Change the heading of the first column to read: "Input Signal."

Paragraph 4.4.2.3 Intermediate Device Operation at Room Temperature: Subparagraph (c) fourth line delete "or voltage and/or resistance as applicable." Add at end of subparagraph (c), "For totalizers a suitable input signal provided by a master test instrument shall be employed in lieu of input capacitance."

Paragraph 4.4.2.4.1 Tank Unit: In the second sentence immediately following "1.0 percent" add "or 0.5 uuf, whichever is greater."

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Paragraph 4.4.2.4.2 Compensator Sensing Unit: In the second sentence immediately following "1.5 percent" add "or 0.5 uuf, whichever is greater."

Paragraph 4.4.2.6 Tank Unit Calibration: Add the following subparagraph:

"4.4.2.6.1 The test specified in 4.4.2.6 shall be conducted on the first and every tenth tank unit thereafter on the contract or order. In the event a tank unit fails to meet the specified requirements, it shall be rejected. When a tank unit failure occurs the next 10 consecutive tank units manufactured shall be subjected to the test specified in 4.4.2.6."

Paragraph 4.4.2.8 Dielectric Strength: Add the following to the last sentence: "and shall not be conducted between points where circuit elements would be damaged by the test voltage."

Paragraph 4.4.3.1 Complete Gage Scale Error at Room Temperature: Delete the last sentence and substitute the following: "The difference between the observed readings and the readings specified shall not exceed 2.0 percent of indication +1.0 percent of full scale indication."

Paragraph 4.4.3.1.1: In the second sentence change "the observed indicator reading" to "the specified indicator readings."

Paragraph 4.4.3.3 Indicator and Intermediate Device Scale Error at Room Temperature: Delete the seventh sentence and substitute the following: "Where applicable, the output signals for the totalizer shall be recorded for each test point."

Paragraph 4.4.3.3.1: In the first sentence delete "precision resistance decade and/or voltage source, as applicable" and substitute the following: "master test instrument."

Paragraph 4.4.3.10.3 Vibration Isolation Suitability - Intermediate Device: Delete the paragraph in its entirety.

Paragraph 4.4.3.10.4.1 Intermediate Device Vibration Resonance: In the fifth sentence delete "1 hour" and substitute "2 hours." Add the following sentence to the paragraph: "The intermediate device shock mount shall be included as a component part of the intermediate device for purposes of this test."

Paragraph 4.4.3.10.4.2 Intermediate Device Vibration Cycling: Delete the paragraph in its entirety.

Paragraph 4.4.3.10.4.3 Tank Unit Resonance: In the seventh sentence delete "1 hour" and substitute "2 hours."

Paragraph 4.4.3.10.4.4 Tank Unit Cycling: Delete the paragraph in its entirety.

Paragraph 4.4.3.10.4.5: Delete the paragraph in its entirety.

Paragraph 4.4.4.1 Vacuum Tube Replacement: Delete the third and fourth sentences and substitute: "Each vacuum tube or plug-in type amplifier employed in the gage system shall be removed individually and replaced consecutively with six tubes or amplifiers of the same type designation taken at random from stock. The difference in reading obtained with the original tube or amplifier and any of the stock replacement tubes or amplifiers shall not exceed 0.3 percent of full scale deflection."

Paragraph 4.4.4.7 Tank Unit Immersion: Delete the eighth sentence and substitute the following: "The gage reading shall be noted after 5 minutes, and the change of indication from the original reading shall not exceed 2 percent of the full scale indication."

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